A Note on the Title

English *power* descends from Old French *poeir*, from Vulgar Latin **potere*, from Latin *posse*, "to be able," contracted from *potis esse*. The adjective *potis*, "able, capable," goes back to the reconstructed Proto-Indo-European root **poti-*, glossed "master, lord, owner, husband." The same root gives Greek *despotes*, "master of the house," which English holds as *despot*.

The word for capability and the word for domination descend from one root, and that root names the master of a household. The title takes *power* in that older sense.

The gloss repays a second reading. *Husband. Master of the house.* The oldest recoverable sense of the word names a position held inside a household. Chapter 8 argues, on evidence assembled independently of any etymology, that the gendered partition is the original template: operational for millennia before race existed as a concept, prototyped within European households, and reverse-engineered into the racial partition in the fifteenth century. The word and the history locate the first extraction in the same room.

That convergence is offered as an observation and carries no argumentative weight. The reconstruction **poti-* is a reconstruction, the semantic range of a Proto-Indo-European root constrains nothing about fifteenth-century Portugal, and every structural claim in this book is established from the historical and quantitative record rather than from philology. The title would mean what it means without it. It is recorded here because it was noticed after the argument was complete, and because a reader who reaches Chapter 8 may find the coincidence worth the moment it costs.

Author's Preface

Before the formal research program, there was an observation layer. The earliest recoverable document in this project's archive is an essay on the Spanish-American War, dated 30 January 2020, cited as prior work in a later paper examining how race, class, and national origin were deployed by American imperialism. That paper traced Frederick Douglass's resignation from his Haitian diplomatic post over a coaling station seizure, Social Darwinism as the ideological justification layer for extraction, and Du Bois's 1915 observation that European democracies had purchased working-class loyalty with wages drawn from colonial labor. There are likely earlier documents, not yet recovered, that extend the archive further back. What the recovered papers share is a consistent structural observation: the same architecture kept appearing across different centuries and different geographies, using race, class, and national origin as instruments to accomplish the same extraction. The formal research program documented in the pages that follow came to describe, in mathematical language, a structure that years of historical reading had already found.

A second line of work ran in parallel to the papers described below, and it began earlier. On 11 May 2023 an argument about the definition of racism was written up under the title *The definition of racism should be changed... no seriously* and published as a public post. It already carried the claims this book formalizes: that racism begins when race begins, that race was constructed to leverage In-group and Out-group sorting, and that a definition which omits systemic oppression omits the mechanism. It was revised through February 2024, by which point it had become *Redefining Racism: A Mathematical and Historical Approach*, whose abstract already names set theory, discrete mathematics, and historical analysis. That document is the ancestor of Chapter 2 and, through it, of this book. The formal program below and the redefinition argument developed alongside each other and converged.

This book is the completion of that formal program, which began as a semester project proposal.

The first document in that program was a short proposal, drafted 6 October 2023, that carried two titles across its drafts, *From Bias to Bytes: A Machine Learning-Driven Analysis of Systemic Racism and Social Inequalities* and *The Calculus of Discrimination: A Mathematical Model for Analyzing Systemic Racism and Social Policies*. Its central bet was that systemic racism could be formalized: that set theory, discrete mathematics, and machine learning algorithms, taken together, could produce a rigorous model of

how policies distribute harm across racial groups. The paper named four theoretical ingredients (Critical Race Theory, cognitive bias research, the McKelvey–Schofield Chaos Theorem, and set-theoretic mathematics) and called for their synthesis into a unified framework. It proposed modified Venn diagrams as a visual apparatus and identified the In-group / Out-group binary as the foundational structural object. It called for empirical validation against real policy data and proposed an automatic bias-detection capability as the ultimate applied output. The mathematics it promised was absent from its pages. It was a research program in proposal form: a correct identification of the problem and the tools, without the derivations that would make those tools coherent.

The second document was empirical. *The Calculus of Injustice* analysed a national dataset of fatal police shootings, comparing group death counts against population denominators and testing how much racial signal the recorded incident features carry. Black people were killed at more than twice the White per-capita rate. The incident features themselves carried little group-identifying signal once the classifier was cross-validated, which set a clear limit on what incident-only data can establish about the decision to use force. The paper cited *From Bias to Bytes* as the prior work that had named the set-theoretic and machine-learning program. The per-capita disparity was that program's first measured empirical shadow.

The next document built the applied capability the proposal had named. *Exploring Bias and Fairness in Language Models Applied to Hiring* evaluated four language models. Its career-level experiment submitted synthetic résumés carrying explicit and inferred racial markers to GPT-4o. Entry-level and executive-level selections held near demographic parity. Mid-level selections went to White candidates in roughly two cases out of three. The recorded selection prompt contained no racial instruction. The disparity appeared at the career stage where advancement compounds. The aggregate marker-category test was not statistically significant; the skew appeared after disaggregation by career level.

The Original Power executes what those earlier papers described. The set-theoretic framework is no longer a proposal: the Elite (E), the Out-group ($O_{\text{racialized}}$), and the Buffer Class (I_{buffer}) are formally defined, their relationships are derived from historical forcing functions, and the extraction function governing them is given a closed mathematical form. The McKelvey–Schofield Chaos Theorem, named in *From Bias to Bytes* and left undeveloped there, is formally integrated into the Agenda-Setter Trap in Chapter 14, where it closes the loop between majority-rule cycling and the psychological wage (ψ) that forecloses the one coordination the theorem identifies as a defense. The modified Venn diagrams of the proposal have become partition theorems, directed spanning-tree proofs, and an electrodynamic circuit formalism that unifies every equation in the book

into a single coherent physical system. The empirical disparity result (the racial gap in fatal-shooting rates, bounded by the limits of incident-only data) is one anchor case in an archive of 146 anchor cases spanning five centuries of the extraction algorithm's operation.

The book carried three titles across its own development, and the sequence records the same progression. It began as *Redefining Racism*, a single argument combining the set-theoretic apparatus with a structural analysis of what racism is; that argument survives as Chapter 2, and the remaining chapters grew outward from it. As the framework generalized past the racial axis, the manuscript became *The Mathematics of Oppression.* It became *The Original Power* when the electrodynamic formalism closed and the object under analysis resolved from a mathematics of a subject into the mechanism itself. The racial axis was the starting point for reasons of measurement. It carries the deepest documentary record and the strongest public datasets, which makes it the axis on which a structural claim can be calibrated before it is transposed. That choice concerns evidence density and states nothing about historical priority. On the question of priority this book takes the opposite position: Chapter 8 establishes the European subjugation of women as the original template, operational for millennia before race existed as a concept, and shows that the racial partition was reverse-engineered from it—legal erasure of personhood, asymmetric autonomy restriction, ideological naturalization, and the weaponization of reproduction were prototyped there first. The formal program calibrates on the younger axis because that is where the instruments read, and transposes to the older one. Appendix I formalizes that transposition and states the invariance condition it requires.

The progression from proposal to empirical detection to applied audit to formal derivation defines the framework's method: identify a structure, detect its empirical shadow, build the instrument that measures it, then derive the mechanism that produces all three. *From Bias to Bytes* identified the structure. *The Calculus of Injustice* detected the shadow. *Exploring Bias and Fairness in Language Models Applied to Hiring* built the detector and turned it on the machine substrate. This book derives the mechanism, and follows that mechanism to its conclusion.

Emmanuel Theodore

Preface

This book develops a formal mathematical framework for analyzing systems of oppression using set theory, discrete mathematics, electrodynamics, and historical analysis. Its central claim is that racism functions as **psycho-legal social software**: legal, institutional, cultural, and affective code that runs on human predictive cognition (the **wetware** substrate on which that software executes) causing engineered partitions to execute as perception, common sense, threat detection, and selective empathy. Running in parallel to this software layer is a **hardware layer**: an electrodynamic formalism that treats the five-tier hierarchy as a circuit topology, the suppression allocation as a complex power signal, and reform shocks as inductive kickback, unifying every equation in the book into a single coherent physical system. Within that hardware layer, the material-wage electric field resolves into a vector superposition of discrete material wages indexed by intersection: a racial material wage, a gendered material wage, a class material wage—each coupling its own axis of privilege to the same extraction circuit. In racism's case, that software is a legally engineered racial partition that exploits ordinary human cognition, recruits predictive shortcuts into racial priors, and turns those priors into a durable extraction system.

A note on epistemology and dynamical homology. The framework establishes a substrate-independent dynamical homology: distinct systems exhibit equivalent governing equations once their state variables and constraints are operationalized. The claim concerns the equations; the specific units are irrelevant to the homology. Both electrodynamic and socioeconomic control architectures are downstream implementations of the same universal dynamical systems equations, forced to optimize energy, labor, and suppression under structural constraints. Every circuit-theoretic, field-theoretic, and control-theoretic statement in this book reads through that homology. Chapter 1 formalizes it as an augmented Lagrangian control system and specifies the falsification tests it must survive.

A critical architectural clarification: the Elite (E) *gate* the energy that drives this system. The kinetic labor, taxes, and physical output of the Out-group and the Buffer Class constitute the actual power supply (V_{cc}) of the extraction circuit. The Elite function as a parasitic control layer—a transistor base receiving only a trickle of control current (laws, algorithms, media narratives) while dictating how the massive power of the masses is routed. The police, courts, and carceral apparatus that enforce the partition are paid

for with wealth extracted from the very populations they patrol. The system is a self-exciting generator: it uses a portion of its own extracted output to power the interference engine that keeps the machine spinning. This means the architecture is powered by *your* compliance, and it dies when that compliance is withdrawn.

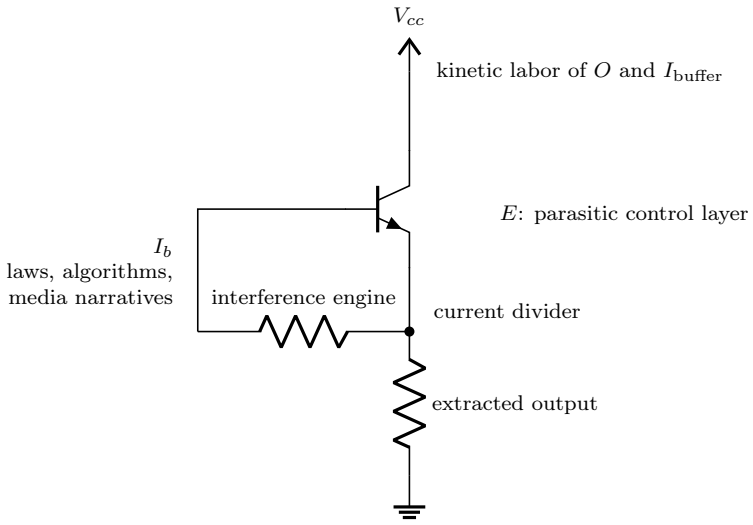


Figure P.1: **The Elite as a Parasitic Control Layer.** Emitter current enters a current divider. The output branch leaves as extracted output. The feedback branch attaches to the base and powers the interference engine.

A note on terms and host wetware. The words **In-group** and **Out-group** are standard social-psychological terms before they are formal variables in this book. In APA Dictionary usage, an ingroup is a group to which a person belongs or with which the person identifies; an outgroup is a group outside the one to which the person belongs or with which the person identifies; and ingroup bias names the tendency to favor one's own group relative to other groups American Psychological Association 2026a; American Psychological Association 2026c; American Psychological Association 2026b. Here, **bias** means a tendency to favor or disfavor people in judgments and evaluations because of group membership; implicit bias operates when past experience shapes those responses outside conscious awareness of its influence Greenwald and Banaji 1995. **Prejudice** is antipathy toward a group or its members grounded in an erroneous generalization held rigidly, following Allport's account of ethnic prejudice Allport 1954, p. 9. **Discrimination** is behavior that treats people differently because of group membership; the three-part distinction identifies a thinking component (stereotype), a feeling component (prejudice), and a behavioral component (discrimination) Fiske 1998. Social Identity Theory then explains why these categories matter: the mind rapidly distinguishes "us" from "them" to

manage trust, threat, cooperation, belonging, and self-concept Tajfel and Turner 1979. In this book's computing analogy, that sorting tendency is part of the human host wetware. The brain is a predictive, pattern-compressing neural network optimized for survival under uncertainty; it prioritizes survival under uncertainty over neutral truth-tracking under adversarial input. What once functioned as useful legacy code for small-group survival becomes a vulnerability when legal and social systems feed it malicious priors. Racism, in this framework, is the exploit: **psycho-legal social software** that installs a racial partition into ordinary predictive cognition and makes the partition feel like perception.

The operating name for that exploit is the **fractal mind virus**. It is "fractal" because the same partition-and-extraction logic reproduces at every scale: the empire, the nation, the city, the school district, the household, the ballot line, and the private reflex of suspicion. It is a "mind virus" because the legal code enters the person. It installs priors into human predictive cognition until the host experiences an engineered hierarchy as common sense. The book therefore tracks two coupled runtimes throughout: institutional code executing through law, property, policing, and finance; and cognitive code executing on human host wetware through fear, status, disgust, loyalty, and selective empathy.

The formal terms used later compress that psychological substrate into structural roles. **In-group** means the population the system marks as relatively protected, legitimate, or entitled to status inside a given partition. **Out-group** means the population marked as extractable, controllable, or disposable. These are structural positions; a person can be advantaged along one axis and subordinated along another. The **Elite** is the small subset of the In-group that actually benefits from the extraction system; the **Buffer Class** is the broader In-group layer recruited to defend the partition in exchange for status, selective protection, or occasional material concessions; the **Puppet Class** translates Elite preference into law and policy; and the **Enforcement Class** physically actuates those policies. The notation introduced later simply lets those recurring roles be tracked precisely.

The framework identifies four architectural components common to oppressive systems: (1) asymmetric autonomy restriction between In-groups and Out-groups, (2) selective empathy that validates In-group suffering while dismissing Out-group harm, (3) ideological justification through spurious claims, and (4) resistance to structural critique. These components operate through a five-tier hierarchy. The Elite (E) extracts value; the Puppet Class (P_{uppet}) translates extraction into law and policy; the Enforcement Class (F_{enforce}) actuates the system physically; the Buffer Class (I_{buffer}) receives a suppression allocation in exchange for policing the partition; and the Out-group ($O_{\text{racialized}}$) bears the compounding burden of extraction.

Before the historical timeline begins, **Chapter 0: System Initialization** compiles the abstract geometry of the trap: the five structural nodes (E, P_{uppet}, F_{enforce}, I_{buffer}, O), their arrangement as a 3-D cone that captures structure the flat 2-D triangle misses, the Tri-Modal Enclosure Model that prevents the Out-group from stepping outside that cone to perceive its own architecture, and the optical illusion—Elite Obscuration and the Flat Ceiling—that makes the apex invisible from the base. Chapter 2 then loads racism as the primary empirical dataset: defining the software before running its execution history. The historical compile begins in Chapter 3 and is instantiated on American wetware in Chapter 4.

This framework did not arrive fully formed. It began as hand-written summations of the familiar two-tier binary—Oppressor versus Oppressed. The equations broke. The model expanded to three tiers to account for the Elite's separation from the broader In-group. Three tiers broke. The law required a Puppet Class; the carceral state required an Enforcement Class. The five-tier architecture is therefore the minimum configuration the mathematics required to remain internally consistent with the historical record.

Reader Diagnostic: Veil of Ignorance and Radical Empathy

As you move through this framework, practice what John Rawls called the **Veil of Ignorance**: evaluate the system from an "original position" in which your tier is hidden from you Rawls 1971. Rawls built this thought experiment on Kantian moral foundations, but its use here is diagnostic, with no prescriptive aim. Its practical discipline is **radical empathy**: a positional stress test in which the reader places themselves inside each tier of the architecture—E, P_{uppet}, F_{enforce}, I_{buffer}, and O—while temporarily stripping away the knowledge of where they actually stand. Radical empathy is a structural modeling discipline. It asks the reader to map the incentives, risks, fears, rewards, and constraints that each tier imposes on the person trapped inside it, and to hold that map alongside the full accounting of harm the architecture produces.

These two instruments serve distinct functions. Radical empathy is the activation step: it asks you to extend the same interpretive charity to a member of any other tier that you extend to yourself—to perceive their constraints as real before evaluating their choices. A reader must first model another tier's incentive structure before the Veil can operate; radical empathy supplies that prerequisite. The Veil is then the computation step: it strips remaining positional information (tier, phenotype, accumulated advantage) and asks you to evaluate the architecture on its structural merits alone. Resistance to the Veil is a system symptom. The fractal mind virus

installs phenotype loyalty as common sense; the Veil exposes that loyalty as a choice, which is why contact with it produces friction. Feeling the resistance identifies the install point.

The diagnostic question is simple: if you had no clue which tier you would wake up in, would you co-sign this architecture? If the answer is no, then welcome to the club. The work is to dismantle the architecture that makes innocence structurally unavailable.

That sequence matters because the book defines racism by tracing the life cycle of an extraction algorithm. **Part I** (*Specification and Origins, 1440s–1915*) establishes the template: the racial vector invented in Zurara's 1453 *Crónica* (Chapter 3), the Buffer Class formalized after Bacon's Rebellion, and the Puppet Class prototyped at the Constitutional Convention (Chapter 4), the constitutional kernel that compiles those prototypes into federal firmware (Chapter 5), and the specification's closing argument in the Haitian export and the Firmin Protocol (Chapter 6). **Part II** (*The Installation, 1619–1865*) builds the on-premise machinery: the colonial extraction of African kinship (Chapter 7), the gendered reproductive kernel of coverture and eugenics (Chapter 8), and the slave-patrol genealogy that compounds into the 13th Amendment loophole (Chapter 9). **Part III** (*Scaling and Runtime, 1865–Present*) shows the algorithm executing at industrial scale: spatial containment and the Capture Variable (Chapter 10), the industrialized Puppet Class and the Tweedism Filter on democracy (Chapter 14), the post-*Brown* Variable Swap and the 1968–1994 recompile (Chapter 15), the terminal runtime of cannibalization and the 5-Tier Reveal (Chapter 16), and the kinetic guarantee that stands between the algorithm and completion (Chapter 18). **Part IV** (*Diagnostics and Output*) produces the framework's terminal findings: the Concession Theorem (Chapter 19), the global containment field (Chapter 20), and the revised, vector-valued definition of racism that closes the book (Chapter 25).

The through-line is expansion. Tracing the algorithm from Portuguese racialization through the invention of whiteness, the 13th Amendment loophole, redlining, the War on Drugs, and the present-day cannibalization of the In-group reveals that the Out-group targeted by systemic oppression *expands* over time, progressively encompassing groups once protected by the In-group boundary. This expansion shows that oppressive systems ultimately serve an Elite class ($E \subset I$) that uses division to prevent solidarity; the nominal In-group receives only derivative protection.

In plain terms, this is the old strategy of divide and conquer rendered as an engineering problem. The set-theoretic model formalizes *how* division is manufactured: the Elite

partitions the population along available axes (race, gender, class, religion, identity) and tunes those partitions to interfere destructively with one another, preventing unified class solidarity from achieving coherent amplitude. The result is a population that can be politically hyperactive while remaining structurally inert: fully engaged in N competing conflicts, none of which touch the extraction kernel. Every chapter traces a different mechanism by which that division is manufactured, maintained, and repaired when it begins to fail. The purpose of the formalism is to make the machine visible enough to disable.

Empirical validation. The formal claims in this book are anchored to reproducible evidence. Every equation and structural assertion is assigned a confidence tier (Tier 1: directly reported or transparently derivable from a peer-reviewed source or public dataset, with no undisclosed analytical step; Tier 2: public dataset with disclosed author operationalisation; Tier 3: ordinal or structural claim with no quantitative calibration attempted and its basis stated) and a falsification criterion. The calibration rests on 146 anchor cases spanning 146 historical events; each case is documented with its data sources, operationalisation procedure, and the conditions under which the structural claim it validates would be falsified. A complete per-equation index (mapping each equation to its confidence tier, primary data source, and falsification criterion) is compiled in the Empirical Validation Index. The Empirical Methodology chapter (p. lxv) details the anchor-and-scale procedure, the sociological precedent for ordinal composite scoring, and the reproducibility standard applied uniformly across all three tiers.

Empirical Methodology

Every numerical claim in this book rests on one of three reproducibility tracks: a peer-reviewed source with a stable identifier, a public dataset whose operationalization is disclosed in full, or an author-constructed estimate whose inputs, transformations, and outputs are stated so that any reader can replicate the calculation from the same starting material. This chapter documents the methodology that governs all four of those tracks: the sociological precedent that legitimises ordinal composite scoring, the anchor-and-scale procedure that grounds the kinetic-resistance variable ρ_τ, the three-tier confidence scheme that labels every quantitative claim, and the reproducibility standard to which every claim is held.

Sociological Estimation Legitimacy

Ordinal composite scoring is the standard instrument of empirical political science and sociology. The datasets and works below each demonstrate that peer-reviewed scholarship publishes ordinal or composite scores as empirical findings; the framework's use of the same methodology is therefore epistemologically legitimate.

Polity IV / Polity5 Marshall, Gurr, and Jaggers 2016. Marshall and Gurr's Center for Systemic Peace assigns every country-year a composite democracy score on a -10 to $+10$ ordinal scale derived from five component indicators; this score is among the most widely cited empirical measures in comparative politics.

V-Dem (Varieties of Democracy) Coppedge et al. 2024. Coppedge et al. produce multi-dimensional democracy indices by aggregating expert-coded ordinal inputs through a Bayesian item-response model; dozens of peer-reviewed articles treat the resulting decimal scores as quantitative findings.

General Social Survey (GSS) NORC at the University of Chicago 2024. NORC at the University of Chicago has fielded a nationally representative longitudinal survey since 1972; the bulk of its social-attitude items are ordinal Likert scales whose aggregates appear as quantitative trend findings in thousands of peer-reviewed articles.

American National Election Studies (ANES) American National Election Studies 2024. The University of Michigan / Stanford ANES Time Series tracks issue-salience rankings and 0–100 feeling-thermometer scores; these ordinal and quasi-interval scales are routinely reported as empirical measures of political opinion.

Correlates of War Singer and Small 2024. Singer and Small's Militarized Interstate Dispute dataset codes conflict events on a five-point ordinal hostility-level scale (from threat to war); the ordinal values are used directly in quantitative analyses of interstate conflict.

World Inequality Database (WID.world) Piketty, Saez, and Zucman 2024. Piketty, Saez, and Zucman's distributional national accounts produce top-decile and top-percentile wealth-share time series; these shares are composite estimates derived from tax records, survey data, and national accounts adjustments—no single authoritative source yields them directly.

Gallup World Poll social capital indices Gallup, Inc. 2024. Gallup aggregates trust, social-network, and volunteering items into composite social-capital scores; these are ordinal proxies used as empirical findings in policy and academic literature alike.

Putnam's *Bowling Alone* Putnam 2000. Putnam measures the decline of American social capital through a composite of fourteen ordinal proxy variables (club membership, voting, survey trust items); the resulting index is presented as a quantitative empirical finding.

Pew Research political polarization studies Pew Research Center 2014. The Pew Research Center tracks ideological consistency and partisan animosity through composite ordinal scales; the resulting polarization scores are cited as empirical measures in peer-reviewed political-science literature.

The argument is straightforward: if top-tier political science and sociology journals routinely publish ordinal composite scores as empirical findings (and they do) then the framework's use of the same methodology to construct the Era-Level Interference Calibration Matrix and the kinetic-resistance variable ρ_τ fits squarely within established scientific practice.

Anchor-and-Scale Methodology

The kinetic-resistance variable ρ_τ is defined on a normalised scale:

$$\rho_\tau \in [0, 1],$$

where $\rho_\tau = 1.0$ denotes a **kinetic threshold breach**—the point at which the Out-group's organised resistance crosses from non-kinetic to kinetic action at a scale that forces a measurable structural response from the dominant system. All other values in the Era-Level Interference Calibration Matrix (Appendix C) are calibrated as fractions of this

maximum.

The scale is anchored by two events, each of which satisfies all conditions for $\rho_\tau = 1.0$ simultaneously.

Anchor Event	**Date**	**Why it anchors the scale**
Bacon's Rebellion W.E.B Du Bois 1935; Morgan 1975	1676	Unambiguous cross-racial armed uprising; forced the Virginia planter class to invent the Buffer Class (I_{buffer}) and enact the Virginia Slave Codes (1705)—a documented, measurable structural response.
Haitian Revolution James 1989; W.E.B Du Bois 1935	1791–1804	The only successful slave revolution in history; produced permanent kernel termination ($\max(t_{\text{post}}) = 0$ locally)—the only documented instance of the Haitian Theorem's strong-form condition being satisfied.

Both events are *unambiguous* kinetic threshold breaches with documented, measurable structural responses. They are the only two events in the 1450–2026 dataset that satisfy all three conditions for $\rho_\tau = 1.0$ (cross-racial or system-wide kinetic mobilisation, documented structural counter-response, and a measurable, durable change in the system's institutional architecture) simultaneously, so their selection is forced by the data.

Every other ρ_τ estimate in the Era-Level Interference Calibration Matrix is calibrated as a fraction of these two anchors. An event that produced partial kinetic mobilisation without a full structural response receives a value below 1.0; an event that produced no kinetic component receives a value near 0. The full calibrated matrix appears in Appendix C.

Confidence Tier Scheme

Every numerical claim in this manuscript is assigned one of three confidence tiers. The one-line definition that appears in the Era-Level Calibration Matrix—*"Tier 1 = directly reported or transparently derivable from a peer-reviewed source or public dataset, with no undisclosed analytical step; Tier 2 = public dataset with disclosed author operationalisation; Tier 3 = ordinal or structural claim with no quantitative calibration attempted and its basis stated"*—is expanded here into a full, operationalisable specification.

Tier 1 — Peer-reviewed quantitative alignment. A Tier 1 claim is calibrated against at least one peer-reviewed source or public dataset carrying a DOI or stable URL; the numerical result is either directly reported in that source or is directly and transparently derivable from its published figures. The reader can verify the number without performing any undisclosed analytical step. *Examples*: Piketty–Saez–Zucman top-decile wealth shares Piketty, T. and Saez, E. and Zucman, G 2018; Gilens–Page policy-responsiveness regression coefficients Gilens, M. and Page, B.I 2014; ACLU cannabis arrest disparity ratios Edwards, Bunting, and Garcia 2013; Edwards, E. and Greytak, E. et al. 2020.

Tier 2 — Public dataset with author computation. A Tier 2 claim uses a publicly accessible dataset, but the author performs the operationalisation and computation; the method is disclosed in the case study or footnote so that a reader possessing the same dataset can reproduce the result. *Examples*: Author-computed ρ_τ intervals derived from Correlates of War hostility-level distributions Singer and Small 2024; Gallup social-capital indices mapped to Φ_{load} ranges Gallup, Inc. 2024.

Tier 3 — Ordinal estimate with method disclosed. A Tier 3 claim is an ordinal ordering or structural relationship for which no quantitative calibration is possible or is attempted. The ordinal basis and its limits are explicitly stated in the text. *Examples*: Global-scaling estimates where regional data is fragmented; pre-1700 era estimates where quantitative records are sparse (see the Global Scaling row in Appendix C).

The tier assignment is the reproducibility standard for each numerical claim. Every equation in the manuscript that carries a ρ_τ or Φ_{load} value is tagged with its tier in the surrounding text. A complete tier-by-tier index of all equations appears in the Empirical Validation Index.

Reproducibility Standard

Every numerical claim in this manuscript satisfies exactly one of three reproducibility tracks:

1. **Peer-reviewed source.** A DOI or stable URL is cited; the claim is directly reported or directly derivable from the cited source without undisclosed analytical steps.

2. **Public dataset with URL.** The dataset's public access URL is cited; the operationalisation method is disclosed in the relevant case study or footnote.

3. **Author-constructed estimate.** The estimation method is disclosed in full (inputs, transformation, and output) so that a reader can replicate the estimate from the same inputs.

Claims not meeting any of these three tracks are flagged as ordinal (Tier 3) and are explicitly marked as such in the text. All Jupyter notebooks used for Tier 1 and Tier 2 computations are available in `Paper/scripts/` and are reproducible via `make empirical`. The per-equation falsification criteria in each case study follow the same framework established in Appendix D.

Part I

Specification and Origins (1440s–1915)